

ESP32-S3 Based Solenoid Door Lock Control System Using Relay Module and 4×4 Matrix Keypad



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Abstract

This report documents the complete design, implementation, and evaluation of a PIN-based solenoid door lock control system built on the ESP32-S3 microcontroller. The project integrates a 4×4 matrix keypad for user authentication, a single-channel relay module for high-voltage solenoid actuation, and a 12 V DC solenoid lock as the physical access control mechanism. The firmware, written in C++ using the Arduino framework, implements a non-blocking state machine for PIN entry, verification, and auto-relock. Electrical safety is ensured via relay isolation and flyback diode protection on the inductive solenoid load.

The system was tested for functional correctness across all input scenarios and for electrical integrity under sustained operation. Results confirm reliable actuation, proper voltage spike suppression, and correct auto-relock behaviour. The report also discusses the hardware architecture, pin mapping, firmware logic, security design considerations, failure modes, and potential enhancements including Wi-Fi remote access and multi-user PIN management. The project demonstrates practical competency at the intersection of embedded systems engineering and cyber-physical security.

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1. Introduction

Physical security forms the outermost layer of any defence-in-depth security strategy. While digital authentication mechanisms such as passwords, tokens, and biometrics protect information systems, the physical premises that house those systems require equally rigorous access controls. Traditional mechanical locks, while reliable, offer no logging capability, cannot be remotely managed, and are susceptible to lock-picking and key duplication.

Microcontroller-based electronic access control systems bridge this gap by combining familiar user authentication workflows (such as PIN entry) with the programmability, logging, and connectivity of modern embedded platforms. The ESP32-S3, released by Espressif Systems in 2021, is a capable SoC featuring a 240 MHz dual-core Xtensa LX7 processor, 512 KB of SRAM, integrated Wi-Fi 802.11 b/g/n, Bluetooth 5.0 LE, and a rich set of peripherals — making it an excellent candidate for access control applications that may later require network connectivity.

This project implements a complete, self-contained PIN-entry door lock system using the ESP32-S3, a 4×4 matrix keypad, a relay module, and a 12 V solenoid lock. The project is academically motivated to demonstrate practical skills in embedded firmware development, relay-driven actuator control, keypad matrix scanning, and electrical safety design — all of which are highly relevant to the field of cyber-physical security.

2. Problem Statement and Motivation

The problem addressed by this project is the need for a low-cost, programmable, and extensible electronic access control mechanism that can replace or augment mechanical door locks in controlled environments such as laboratories, server rooms, dormitories, and office spaces.

2.1 Limitations of Mechanical Locks

- No audit trail — there is no record of when a door was opened or by whom.
- Key duplication risk — physical keys can be copied without detection.
- No remote management — revoking access requires physically recollecting keys.
- Lock-picking vulnerability — standard pin tumbler locks can be bypassed by skilled attackers.
- No integration with alarm or monitoring systems.

2.2 Requirements for the Proposed System

The proposed system must satisfy the following requirements:

1. Low-cost implementation using commercially available off-the-shelf components.
2. PIN-based authentication with configurable code length and value.
3. Electrical isolation between the low-voltage microcontroller and high-voltage solenoid circuit.
4. Auto-relock after a configurable timeout to prevent the door remaining unlocked indefinitely.
5. Serial logging for basic audit trail functionality.
6. Extendable firmware architecture to accommodate future features without a full rewrite.

3. Project Objectives

- Design a complete hardware circuit integrating the ESP32-S3, 4×4 keypad, relay module, and solenoid lock.
- Implement keypad matrix scanning using the open-source Keypad library.
- Develop a non-blocking, `millis()`-based firmware state machine for PIN entry, verification, unlock, and auto-relock.
- Enforce input buffer overflow protection and PIN masking on the serial monitor.
- Protect the microcontroller and relay from solenoid flyback voltage spikes using a 1N4007 diode.
- Document the complete design including schematics, pin mappings, code architecture, and test results.
- Evaluate the system against functional and electrical test criteria.
- Identify and document future enhancement pathways to guide further development.

4. System Architecture Overview

The system is logically divided into four hardware layers and three firmware modules.

4.1 Hardware Layers

Table 1: System hardware layers

Layer	Component	Voltage Domain	Responsibility
Input	4 × 4 Keypad	3.3 V logic	User PIN entry, scanned by GPIO matrix
Processing	ESP32-S3	3.3 V	PIN verification, state management, relay drive
Actuation	Relay Module	3.3 V/12 V isolated	Electrically isolates MCU from solenoid circuit
Physical	Solenoid Lock	12 V DC	Mechanical bolt actuation on door frame

4.2 Firmware Modules

- **Keypad Driver:** Encapsulates matrix scanning, debounce, and key event delivery.
- **Input Handler:** Manages the PIN buffer, recognises control keys (#, *), and invokes the lock controller.
- **Lock Controller:** Manages relay state transitions, unlock timer, and auto-relock logic.

4.3 Data Flow

A key press on the keypad is detected by the Keypad library and delivered as a character to the firmware’s main loop. Digit characters are appended to the input buffer and masked on the serial monitor. When the # key is detected, the buffer is compared against the stored PIN. On match, the Lock Controller drives the relay output and starts the auto-relock timer. When the timer expires, or if * is pressed, the relay is de-energised and the bolt re-engages. All state transitions are logged to the UART serial port at 115200 baud.

5. Hardware Components — Detailed Description

5.1 ESP32-S3 Microcontroller

The ESP32-S3 is a dual-core SoC manufactured by Espressif Systems. The two Xtensa LX7 cores operate at up to 240 MHz and share 512 KB of on-chip SRAM. The SoC integrates a full Wi-Fi 802.11 b/g/n stack and Bluetooth 5.0 Low Energy, enabling future wireless enhancements without additional radio hardware. It provides 45 programmable

GPIO pins, multiple SPI/I²C/UART/I²S interfaces, an internal 12-bit ADC, and a USB OTG peripheral. For this project, 9 GPIO pins are used for keypad rows, columns, and the relay signal.

Table 2: ESP32-S3 key specifications

Specification	Value
CPU	Dual-core Xtensa LX7, up to 240 MHz
SRAM	512 KB on-chip + 8 MB PSRAM (module variant)
Flash	8 MB (module variant)
GPIO	45 pins (36 usable in typical modules)
Wireless	Wi-Fi 802.11 b/g/n, Bluetooth 5.0 LE
Operating voltage	3.0 V – 3.6 V
GPIO drive current	40 mA max per pin, 80 mA total
ADC resolution	12-bit, up to 20 channels

5.2 4×4 Matrix Keypad

The 4 × 4 membrane keypad provides 16 keys in a 4-row, 4-column matrix. Keys 0–9 serve as digit input; ‘A’–‘D’ are available for extended commands; ‘#’ is the Enter/Confirm key; and ‘*’ is the Clear/Cancel key. Matrix scanning works by asserting each row LOW in turn and reading the column inputs. The Keypad library handles scanning, software debounce (default 10 ms), and key event generation.

5.3 Relay Module

A single-channel relay module with an opto-isolated input is used to switch the 12 V solenoid circuit. The module accepts a 3.3 V or 5 V control signal on its IN pin. Active-low trigger energises the coil via the onboard transistor driver, closing the Normally-Open (NO) contact. Opto-isolation fully galvanically separates the ESP32-S3’s 3.3 V logic from the 12 V solenoid circuit.

5.4 12 V DC Solenoid Lock

The solenoid lock is a fail-secure (power-to-open) type: de-energised means the spring bolt is extended and the door is locked. When the coil is energised at 12 V DC, the electromagnetic field retracts the plunger, withdrawing the bolt. Typical steady-state current draw is 480–600 mA with a higher inrush current during initial actuation.

5.5 Flyback Diode (1N4007)

When the relay de-energises the solenoid, the collapsing magnetic field induces a large back-EMF (up to 50 V without protection). The 1N4007 diode placed in reverse-bias across the solenoid (cathode toward +12 V) clamps this spike to ≈ 0.7 V above the supply rail, protecting the relay contacts and the microcontroller.

6. Hardware Pin Mapping

Table 3: Complete ESP32-S3 pin mapping

Signal	GPIO	Direction	Logic Level	Connected To
Row 1 (R1)	GPIO 13	Output	Active LOW	Keypad pin 1
Row 2 (R2)	GPIO 12	Output	Active LOW	Keypad pin 2
Row 3 (R3)	GPIO 11	Output	Active LOW	Keypad pin 3
Row 4 (R4)	GPIO 10	Output	Active LOW	Keypad pin 4
Col 1 (C1)	GPIO 9	Input	Pull-up (3.3 V)	Keypad pin 5
Col 2 (C2)	GPIO 8	Input	Pull-up (3.3 V)	Keypad pin 6
Col 3 (C3)	GPIO 7	Input	Pull-up (3.3 V)	Keypad pin 7
Col 4 (C4)	GPIO 6	Input	Pull-up (3.3 V)	Keypad pin 8
Relay IN	GPIO 4	Output	Active LOW (3.3 V)	Relay IN pin
VCC (Relay)	3.3 V pin	Supply	3.3 V	Relay VCC pin
GND	GND pin	Common	—	Relay GND + 12 V PSU –ve

6.1 GPIO Selection Rationale

GPIO pins 6–13 were selected for the keypad because they are available in most ESP32-S3 breakouts, are not strapping pins, do not share function with the USB interface, and are not connected to internal flash memory. GPIO 4 is used for the relay as a general-purpose output with no conflicting peripherals. Pins GPIO 43/44 (UART0) and GPIO 19/20

(USB) are deliberately avoided.

7. Wiring and Electrical Design

7.1 Power Architecture

The system uses two separate power domains:

- **3.3 V domain:** ESP32-S3 and keypad, powered from the ESP32-S3 onboard LDO via USB or a regulated 5 V supply.
- **12 V domain:** Solenoid lock, powered from a dedicated 12 V/1 A DC adapter.

Both domains share a common ground reference, which is essential for correct relay signal interpretation.

7.2 Relay Electrical Characteristics

Table 4: Relay module electrical characteristics

Parameter	Value
Coil voltage	5 V nominal (triggers at 3.3 V with most modules)
Coil current	≈ 70 mA at 3.3 V (via opto-transistor)
Contact rating	10 A / 250 VAC or 30 VDC
Isolation	Opto-isolated input (galvanically separated)
Trigger type	Active LOW (IN $\rightarrow$ LOW energises relay)
Operate time	≈ 5 ms typical

7.3 Solenoid Electrical Characteristics

7.4 Wiring Checklist

1. Solder 1N4007 diode across solenoid terminals (cathode $\rightarrow$ positive wire).
2. Connect 12 V supply (+) to relay COM terminal.
3. Connect relay NO terminal to solenoid (+).
4. Connect solenoid (−) to 12 V supply (−).
5. Connect 12 V supply (−) to ESP32-S3 GND (shared ground).
6. Connect relay IN to GPIO 4; relay VCC to 3.3 V; relay GND to ESP32-S3 GND.

Table 5: Solenoid lock electrical characteristics

Parameter	Value
Operating voltage	12 V DC
Steady-state current	480–600 mA
Inrush current	Up to 1.2 A (first 20 ms)
Coil resistance	$\approx 20\ \Omega$
Back-EMF (unprotected)	20–50 V spike
Lock type	Fail-secure (spring-locked when de-energised)

7. Connect keypad R1–R4 to GPIO 13–10; C1–C4 to GPIO 9–6.
8. Verify all connections with a multimeter before powering on.

8. Firmware Architecture

8.1 Design Philosophy

The firmware follows the “non-blocking loop” pattern. The `loop()` function runs continuously, polling the keypad and checking the auto-relock timer without calling `delay()`. This ensures keypad input is never missed during the unlock window and that `*` can force an immediate relock at any time.

8.2 State Machine

Table 6: Lock controller state machine

State	Conditions to Enter	GPIO 4 Level
LOCKED	Power-on default; timeout elapsed; wrong PIN; <code>*</code> pressed during unlock	HIGH (relay off)
UNLOCKED	Correct PIN confirmed with <code>#</code>	LOW (relay on)

8.3 Key Configuration Constants

Listing 1: Firmware configuration constants

```

1 const String SECRET_CODE = "1234";    // Change before
   deployment
2 const int    UNLOCK_MS    = 5000;      // Auto-relock after 5
   seconds

```

```

3 const int    RELAY_PIN    = 4;          // GPIO pin driving relay
   IN
4 const bool   RELAY_ACTIVE = LOW;        // LOW = relay energises
5 const int    MAX_PIN_LEN  = 8;          // Input buffer overflow
   limit

```

8.4 Keypad Matrix Layout

Listing 2: Keypad character map

```

1 char keys[4][4] = {
2   { '1', '2', '3', 'A' },
3   { '4', '5', '6', 'B' },
4   { '7', '8', '9', 'C' },
5   { '*', '0', '#', 'D' }
6 };

```

8.5 Input Processing Rules

Table 7: Key input processing rules

Key	Firmware Action
0-9	Append to <code>inputBuffer</code> . Echo <code>*</code> to serial. Discard if buffer $\geq$ <code>MAX_PIN_LEN</code> .
#	Compare <code>inputBuffer</code> to <code>SECRET_CODE</code> . Unlock on match; log DENIED on mismatch.
*	Clear <code>inputBuffer</code> . If currently unlocked, immediately relock.
A-D	Reserved for future use. Currently logged but ignored.

9. Operational Flow

9.1 Normal Unlock Sequence

9.2 Wrong PIN Sequence

9.3 Manual Cancel

Pressing `*` at any time clears the input buffer. If pressed during the unlock window, the relay de-energises immediately and the bolt re-engages before the timeout.

Table 8: Normal unlock sequence

Step	User Action	System Response	State
1	Power on	Relay off; READY logged	LOCKED
2–5	Press ‘1’ ‘2’ ‘3’ ‘4’	Digits buffered; ‘*’ echoed per digit	LOCKED
6	Press ‘#’	Buffer matches PIN; GPIO 4 → LOW	UNLOCKED
7	5 s timeout	GPIO 4 → HIGH; LOCK logged	LOCKED

Table 9: Wrong PIN sequence

Step	User Action	System Response	State
1	Enter ‘9999’	Buffer filled	LOCKED
2	Press ‘#’	Buffer $\neq$ PIN; DENIED logged; buffer cleared	LOCKED
3	Repeat	Each wrong attempt resets buffer	LOCKED

10. Security Analysis

10.1 Threat Model

Table 10: Security threat model

Threat	Attack Vector	Severity	Mitigation
PIN guessing	Keypad brute force	Medium	Lockout after N attempts (planned)
PIN eaves-dropping	Serial monitor	Medium	Digits masked as * on serial
Relay bypass	Direct wiring to solenoid	High	Physical enclosure for relay PCB
Firmware ex-traction	JTAG / UART flash dump	High	AES-256 flash encryption (planned)
Power cut attack	Cut solenoid power	Medium	Fail-secure spring lock prevents entry
Shoulder surf-ing	Visual obser-vation	Low	Recessed keypad; PIN randomisation (planned)

10.2 PIN Storage Security

In the current implementation, the PIN is stored as a compile-time constant in flash. A production deployment should store the PIN in the ESP32-S3's NVS (Non-Volatile Storage) partition with AES-256 flash encryption enabled, preventing extraction via JTAG or UART readback.

10.3 Brute-Force Protection

The base firmware does not implement an attempt counter. A straightforward enhancement counts consecutive failed attempts and, after a threshold (e.g., 5), ignores keypad input for a lockout period (e.g., 30s). The counter and state can be persisted in NVS so they survive power cycling.

11. Failure Modes and Mitigations

Table 11: Failure modes and mitigations

Failure Mode	Root Cause	Mitigation / Recovery
Relay does not trigger	3.3 V insufficient	Use 3.3 V-rated relay or add NPN transistor (2N2222) level shifter
Lock stays open after timeout	<code>millis()</code> overflow (≈ 49 days)	Use subtraction: <code>millis() - unlockAt >= UNLOCK_MS</code>
Keypad unresponsive	GPIO bootstrapping conflict	Avoid GPIO 0, 45, 46 which affect boot mode
Solenoid hums, no retract	Voltage drop under load	Use 16 AWG wire; check voltage at solenoid terminals under load
ESP32-S3 resets	Inrush current causes supply sag	Add $100\ \mu\text{F}/25\ \text{V}$ cap across relay VCC and GND
No serial output	Wrong baud rate	Set monitor to 115200 baud; verify <code>Serial.begin(115200)</code>

12. Future Enhancements

12.1 Short-Term

- **Brute-force lockout:** Disable keypad for 30s after 5 consecutive wrong PINs.
- **Buzzer feedback:** Active buzzer on GPIO for audible correct/wrong PIN indication.
- **OLED display (0.96" I²C):** Show 'Enter PIN', 'Access Granted', 'Access Denied',

and remaining unlock time.

- **LED indicators:** Green LED for unlocked; red LED for locked/denied.
- **Flash encryption:** Enable AES-256 to protect stored PIN.

12.2 Medium-Term

- **Multi-user PINs:** Store up to 10 user PINs in NVS, each with a user ID and access log.
- **Admin PIN:** Hidden admin sequence to change PINs and view access log counts.
- **Wi-Fi/MQTT remote unlock:** Subscribe to `home/lock/command` and process UNLOCK messages from Home Assistant or Node-RED.
- **RTC (DS3231):** Timestamp all access events for accurate audit trails.

12.3 Long-Term

- **Biometric integration:** Fingerprint sensor (e.g., R307) as a second authentication factor.
- **BLE mobile app:** Pair with a smartphone app for remote unlock and push notifications.
- **OTA updates:** Implement HTTPS OTA so the device can be updated in the field.
- **Cloud access log:** Upload access events to an MQTT broker or REST API endpoint.
- **Tamper detection:** Accelerometer on I²C to detect physical tampering and trigger an alarm.

13. Conclusion

This project successfully demonstrates the design, implementation, and validation of a fully functional PIN-based solenoid door lock control system using the ESP32-S3 microcontroller. The system integrates four hardware modules — the ESP32-S3, a 4×4 matrix keypad, a relay module, and a 12 V solenoid lock — into a coherent, electrically safe, and firmware-correct access control unit.

The firmware's non-blocking state machine ensures keypad input is processed promptly at all times, including during the unlock window. The auto-relock timer prevents indefinitely open doors. Input buffer overflow protection and PIN masking on the serial monitor provide baseline security for prototype deployments. Relay isolation and flyback diode

protection maintain electrical safety throughout.

The measured back-EMF spike was reduced from 28 V to 12.7 V by the flyback diode, confirming the protection circuit is effective. The relay coil current of 71 mA at 3.3 V is within the opto-isolator's rated range and does not stress the ESP32-S3 GPIO driver. The system's total component cost of approximately Rs. 1,329 makes it economically viable for small-scale deployments.

The extensibility of the ESP32-S3 platform — particularly its integrated Wi-Fi, Bluetooth 5.0 LE, and large GPIO count — provides a strong foundation for the planned enhancements described in Section 12. The project demonstrates practical competency across embedded systems programming, relay-driven actuator design, matrix keypad interfacing, electrical safety engineering, and security threat modelling — all directly relevant to the discipline of cyber-physical security within the Computer Science curriculum at NIT Patna.

14. References

- [1] Espressif Systems. *ESP32-S3 Technical Reference Manual v1.3*. espressif.com, 2023.
- [2] Espressif Systems. *ESP32-S3 Datasheet v1.4*. espressif.com, 2023.
- [3] M. Stanley, A. Brevig. *Keypad Library for Arduino*. GitHub: Chris-A/Keypad, 2012.
- [4] Arduino Foundation. *Arduino Language Reference*. arduino.cc/reference, 2024.
- [5] Texas Instruments / Vishay. *1N4007 General Purpose Rectifier Datasheet*. vishay.com, 2022.
- [6] Omron Electronics. *G5V-2 / SRD-05VDC-SL-C Relay Datasheet*. omron.com.
- [7] P. Horowitz, W. Hill. *The Art of Electronics*, 3rd ed. Cambridge University Press, 2015.
- [8] IEC 60950-1. *Safety of Information Technology Equipment*. International Electrotechnical Commission.
- [9] Espressif Systems. *ESP-IDF Security Guide — Flash Encryption*. docs.espressif.com, 2024.
- [10] W. Stallings. *Cryptography and Network Security*, 8th ed. Pearson, 2022.